Permanent Magnet Materials Thermodynamic Database
Technical Information

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Version 2026a

Contents

Permanent Magnet Materials Database（RRPMAG）

The Permanent Magnet Materials Database (PMAG) is a thermodynamic and property database for rare-earth permanent magnet applications. It is applicable from pure $Nd_2Fe_{14}B$ to complex commercial NdFeB-based and SmCo-based permanent magnet materials. This database can be used not only to calculate phase diagrams and thermodynamic properties of assessed systems, but also to predict phase equilibria and Curie temperatures over wide composition ranges and to simulate solidification processes.

Based on magnet composition and temperature, this database can predict the fundamental thermodynamic behavior and phase equilibria of these systems. These calculations can accelerate the design of new NdFeB-based permanent magnet alloys and improve understanding of their processing and service behavior.

The database was developed using the CALPHAD method and was established on the basis of various experimental data and theoretical values, such as first-principles calculation results. Its development is based on rigorous assessments of binary, ternary, and selected higher-order systems that are important for the database. As a result, predictions can be made for multicomponent systems and alloys of industrial importance. Among these data, the thermodynamic database plays a fundamental role.

Interoperability with Other Products

The Rare-Earth Permanent Magnet Materials Database (RRPMAG) can be used together with the company's ICALPHAD phase-diagram thermodynamic calculation software.

Application Case Examples

Please refer to the RRPMAG1.0 Calculation Examples manual. This manual provides several examples of the database, demonstrates database validation results, and illustrates the types of calculations that can be performed for different materials or application fields.

RRPMAG can predict multicomponent systems and alloys of industrial importance, including multicomponent phase-equilibrium calculations, equilibrium solidification simulations, and Scheil solidification simulations. This means that the database can be extrapolated to higher-order systems by combining multiple rigorously assessed systems.

Depending on the actual alloy composition, this database can be used to calculate the following:

- Phase-based properties: critical transformation temperatures, such as precipitate solvus temperatures, phase fractions and phase compositions, solubility limits, activities, phase diagrams, etc.
- Equilibrium and nonequilibrium solidification: liquidus, solidus, incipient melting temperature, solidification temperature range, solid fraction curve, solidification path, eutectic fraction, microsegregation, partition coefficients, latent heat, shrinkage, etc.
- Curie temperature.

Elements

Al	B	Ce	Co	Cu	Dy	Fe	Ga
La	Nb	Nd	Pr	Sm	Tb	Ti	Zr

It covers the Nd-Fe-B main phase ($RE_2Fe_{14}B$), rare-earth-rich phases/liquid phases, grain-boundary phases such as $Nd_6Fe_{13}X$, Co-related temperature-adjustment systems, Cu/Al/Ga grain-boundary diffusion systems, and competing phases induced by Nb/Ti microalloying. It supports thermodynamic and phase-equilibrium calculations throughout solidification, solid solution treatment, annealing, and grain-boundary diffusion processes.

Developed using the CALPHAD method, it integrates multiple types of experimental and theoretical data, such as first-principles calculations. Binary, ternary, and selected higher-order systems important to the database are rigorously assessed, enabling predictions for multicomponent systems and industrial alloys. The thermodynamic database is of fundamental importance in this framework.

All stable solution phases and intermetallic compounds present in each assessed system have been included. In most cases, phases with the same crystal structure are merged into a single phase.

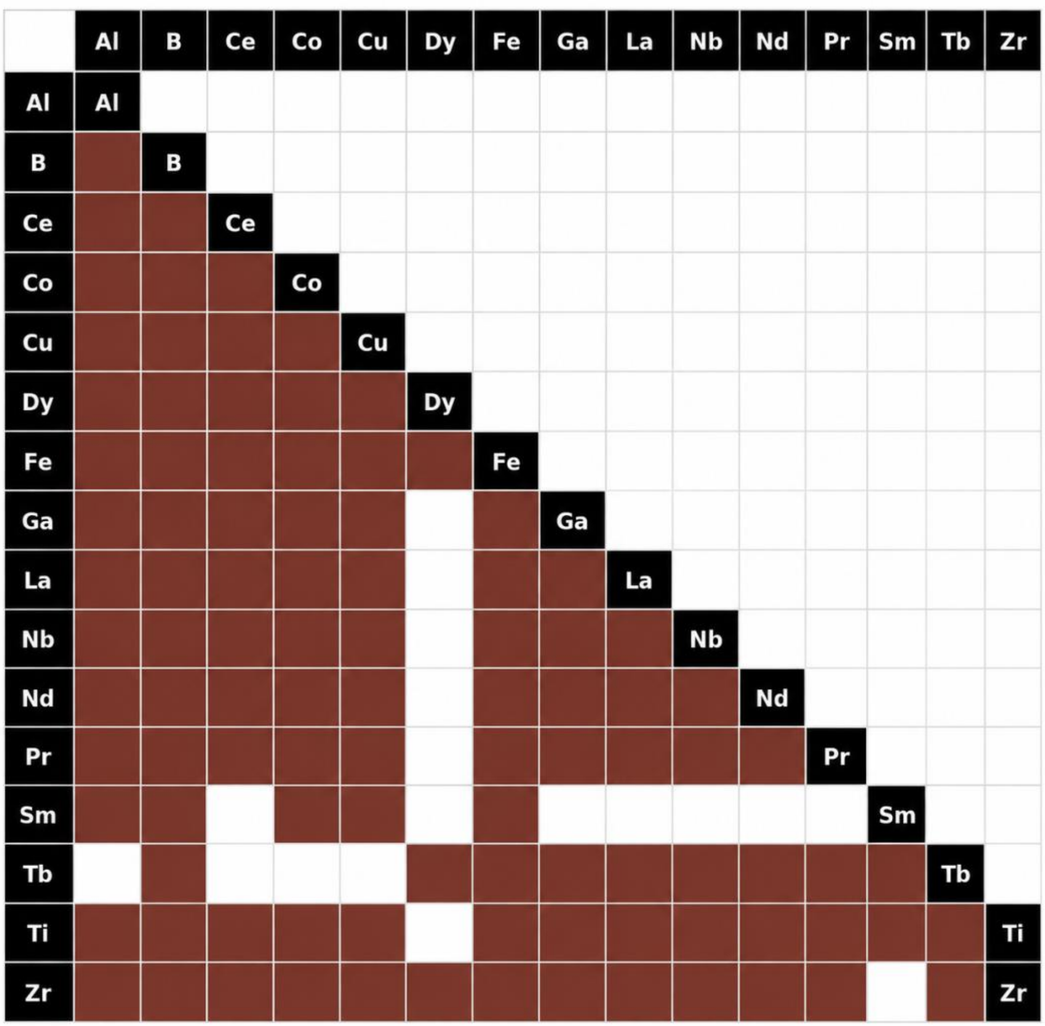
Systems

This database combines experiments, first-principles calculations, and CALPHAD modeling to establish thermodynamic descriptions of binary and ternary systems over the full composition and temperature ranges.

All stable solution phases and intermetallic compounds present in each assessed system are included, and phases with the same crystal structure are merged into the same phase.

Binary Systems

The following figure shows the assessed binary-system matrix in the latest version of the database. The colored cells in the matrix indicate assessed binary systems, totaling 102 systems.



Ternary Systems

The assessed ternary systems in Version 2026a of this database are listed below, totaling 30 systems.

Al-Ce-Co	Al-Ce-Cu	Al-Co-Fe	Al-Co-Ti	Al-Cu-Dy
Al-Cu-Fe	Al-Cu-Nd	Al-Fe-Nb	Al-Fe-Nd	B-Ce-Fe
B-Co-Fe	B-Co-Ti	B-Cu-Fe	B-Cu-Ti	B-Dy-Fe
B-Fe-La	B-Fe-Nb	B-Fe-Nd	B-Fe-Pr	B-Fe-Sm
B-Fe-Tb	B-Fe-Ti	B-Fe-Zr	Ce-Co-Fe	Ce-Fe-La
Ce-Fe-Nd	Co-Fe-Nb	Dy-Fe-Tb	Fe-La-Pr	Fe-Nd-Pr

Phases

Common Phases in Nd-Fe-B Permanent Magnet Alloys

Liquid	Liquid phase used to describe Nd-Fe-B permanent magnet alloy melts, covering the composition $(Al, B, Ce, Co, Cu, Dy, Fe, Ga, La, Nb, Nd, Pr, Tb, Ti)_1$.
T1	$RE_2Fe_{14}B$-based phase, the main hard magnetic phase, covering the composition $(Dy, Nd, La, Ce, Pr, Tb)_2(Fe, Co)_{14}(B)_1$.
T2	$REFe_4B_4$-based phase used to describe the boron-rich $REFe_4B_4$ phase family in the Nd-Fe-B system, covering the composition $(Dy, Nd, La, Ce, Pr, Tb)_1(Fe, Co)_4(B)_4$.
Fe_2B	Fe_2B boride phase, covering the composition $(Fe, Co)_2(B)_2$.
FeB	FeB boride phase, covering the composition $(Fe, Co)_1(B)_1$.
DHCP	Rare-earth-rich phase (double hexagonal close-packed/rare-earth metal phase), covering the composition $(Ce, La, Nd, Pr, Dy, Tb)_1$.
Fe_2RE	Fe_2RE Laves phase used to describe 2:1 rare-earth-iron compounds in the (Ce, La, Nd, Pr, Dy, Tb)-Fe systems, covering the composition $(Ce, La, Nd, Pr, Dy, Tb)_1(Al, Co, Fe)_2$.
ReB_6	RE B_6-type rare-earth boride phase, covering the composition $(Ce, La, Nd, Pr, Dy, Tb)_1(B)_6$.
$ReCo_5$	$RECo_5$-type rare-earth-cobalt phase, covering the composition $(Ce, La, Nd, Pr, Dy, Tb)_1(Co, Fe)_5$.
$ReCo_{12}B_6$	$RECo_{12}B_6$-type rare-earth-cobalt-boron phase, covering the composition $(Ce, La, Nd, Pr, Dy, Tb)_1(Co, Fe)_{12}(B)_6$.
NbB	Nb/Ti boride-type phase used to describe grain-refining/additive phases based on Nb and Ti, covering the composition $(Nb, Ti)_1(B)_1$.

Models Used for Each Phase

The models used for each phase in Version 2026a of this database are listed below, totaling 203 models.

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
$Al_{10}CeFe_2$	$YbFe_2Al_{10}$	-	oS52	(63, Cmcm)	$(Al)_{10}(Ce, Nd)_1(Fe)_2$
$Al_{11}R_3$_HT	Al_4Ba (D_{13})	D_{13}	tI10	(139,I4/mmm)	$(Al)_{11}(La, Nd, Pr, Sm)_3$
$Al_{11}R_3$_LT	$Al_{11}La_3$	-	oI28	(71, Immm)	$(Al)_{11}(Ce, La, Nd, Pr, Sm, Tb)_3$
$Al_{13}Co_4$	Orthorhombic Co_4Al_{13}	-	oP102	(31, Pmn21)	$(Al)_{13}(Co)_4$
$Al_{13}Fe_4$	$Al_{13}Fe_4$	-	mS102	(12, C2/m)	$(Al, Cu)_{0.6275}(Fe)_{0.235}(Al, VA)_{0.1375}$
Al_2Cu_OMEGA	Khatyrkite (Al_2Cu, C_{16})	C_{16}	tI12	(140, I4/mcm)	$(Al)_2(Cu)_1$
Al_2Fe_1	Al_2Fe	-	aP18	(1, P1)	$(Al, Cu)_2(Fe)_1$
Al_2R_3	Zr_3Al_2	-	tP20	(136, P4_2/mnm)	$(Al)_2(Dy, Tb)_3$
Al_2Zr_3_TP20	Zr_3Al_2	-	tP20	(136, P4_2/mnm)	$(Al)_2(Zr)_3$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
Al_3CeCu	Al_4Ba (D_{13})	D_{13}	tI10	(139,I4/mmm)	$(Al, Cu)_4(Ce)_1$
Al_3Ce_H	Unknown Structure	-	-	-	$(Al)_3(Ce)_1$
Al_3Co	Os_4Al_{13}	-	mS34	(12, C2/m)	$(Al)_3(Co)_1$
AlM_3_A_{15}	Cr_3Si (A_{15})	A_{15}	cP8	(223, Pm-3n)	$(Al, Nb)_1(Al, Nb)_3$
Al_3Ce_L	Ni_3Sn (D_{019})	D_{019}	hP8	(194, P6_3/mmc)	$(Al)_{0.75}(Ce)_{0.25}$
Al_3La	Ni_3Sn (D_{019})	D_{019}	hP8	(194, P6_3/mmc)	$(Al)_3(La, Sm)_1$
Al_3Nd	Ni_3Sn (D_{019})	D_{019}	hP8	(194, P6_3/mmc)	$(Al)_3(Nd)_1$
Al_3Pr	Ni_3Sn (D_{019})	D_{019}	hP8	(194, P6_3/mmc)	$(Al)_3(Pr)_1$
Al_3R	Al_3HO	-	hR20	(166, R-3m)	$(Al)_3(Dy)_1$
Al_3Zr_2_OF_{40}	Zr_2Al_3	-	oF40	(43, Fdd2)	$(Al, Ga)_3(Zr)_2$
Al_3Zr_5_D_8M	W_5Si_3 (D_8m)	D_8m	tI32	(140, I4/mcm)	$(Al)_3(Zr)_5$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
$Al_3Zr\_D0_{23}$	Al_3Zr ($D0_{23}$)	$D0_{23}$	tI16	(139,I4/mmm)	$(Al, Ga)_3(Zr)_1$
Al_4Fe	AlmFe	-	tI110	(121, I-42m)	$(Al)_{4.2}(Fe)_1$
Al_4Ce_1	Al_4Ba ($D1_3$)	$D1_3$	tI10	(139,I4/mmm)	$(Al)_4(Ce)_1$
$Al_4Sm\_D_1B$	Al_4U (D_1b)	D_1b	oI20	(74, Imma)	$(Al)_4(Sm)_1$
Al_4Zr_5	Ti_5Ga_4	-	hP18	(193, P6_3/mcm)	$(Al, Ga)_4(Zr)_5$
$Al_{53}La_{22}$	Hexagonal omega (C_{32})	C_{32}	hP3	(191,P6/mmm)	$(Al)_{53}(La)_{22}$
Al_5Co_2	Co_2Al_5 ($D8_{11}$)	$D8_{11}$	hP28	(194, P6_3/mmc)	$(Al)_5(Al, Co)_2$
Al_5Fe_2	$Al_{2.8}Fe$	-	oS24	(63, Cmcm)	$(Al, Cu)_5(Fe)_2$
$Al_{62}Cu_{25}Fe_{13}$	Quasicrystal	-	-	-	$(Fe)_{13}(Al, Cu)_{25}(Al)_7$
Al_7Cu_2Fe	$FeCu_2Al_7$ (E_9a)	E_9a	tP40	(128, P4/mnc)	$(Fe)_1(Cu)_2(Al)_7$
Al_8CeFe_2	$CeFe_2Al_8$	-	oP44	(55, Pbam)	$(Al)_8(Ce)_1(Fe)_2$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
Al_8CeM_4	$Mn_{12}Th$ (D_2b)	D_2b	tI26	(139,I4/mmm)	$(Al)_8(Ce, Nd)_1(Al, Cu, Fe)_4$
Al_8Fe_5	gamma-brass (Cu_5Zn_8, D_{82})	D_{82}	cI52	(217, I-43m)	$(Al, Cu, Fe)_1$
Al_9Co_2	Co_2Al_9 (D_8d)	D_8d	mP22	(14, P2_1/c)	$(Al)_9(Co)_2$
$Al_{10}Cu_{10}Fe$	($Al_{10}Cu_{10}Fe$)	-	oF116	(42, Fmm2)	$(Fe)_1(Al, Cu)_{10}(Al)_{10}$
AlB_{12}	alpha-AlB_{12}	-	tP216	(92, P4_12_12)	$(Al)_1(B)_{12}$
AlB_2_C_{32}	Hexagonal omega (C_{32})	C_{32}	hP3	(191,P6/mmm)	$(Al, Nb, Zr)_1(B)_2$
$REGa_2$_C_{32}	Hexagonal omega (C_{32})	C_{32}	hP3	(191,P6/mmm)	$(Ce, Dy, Ga, La, Nd, Pr, Tb)_1(Ga)_2$
Cu_2La_1	Hexagonal omega (C_{32})	C_{32}	hP3	(191,P6/mmm)	$(Cu)_2(La)_1$
REB_2	Hexagonal omega (C_{32})	C_{32}	hP3	(191,P6/mmm)	$(Dy, Tb)_1(B)_2$
$AlCe_2Cu_2$	Unknown Structure	-	-	-	$(Al)_1(Ce)_2(Cu)_2$
AlCeFe	Unknown Structure	-	-	-	$(Al, Fe)_2(Ce, Nd, Pr)_1$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
AlCeM	Revised Fe_2P (C_{22})	C_{22}(II)	hP9	(189, P-62m)	$(Al)_1(Ce)_1(Cu)_1$
AlCe_OC_{16}	AlCe	-	oS16	(63, Cmcm)	$(Al)_1(Ce)_1$
AlCu_DEL	Al_5Cu_8	-	hR52	(160, R3m)	$(Al)_2(Cu, Fe)_3$
AlCu_EPS	Ni_2In (B_{82})	B_{82}	hP6	(194, P6_3/mmc)	$(Al, Cu)_1(Cu, Fe)_1$
AlCu_ETA	AlCu(r)	-	mS20	(12, C2/m)	$(Al, Cu)_1(Cu, Fe)_1$
AlCu_ZETA	Al_9Cu_{11}(h)	-	oF88	(42, Fmm2)	$(Al)_9(Cu, Fe)_{11}$
Al_3Dy_D_{024}	Ni_3Ti (D_{024})	D_{024}	hP16	(194, P6_3/mmc)	$(Al)_3(Dy, Tb)_1$
Al_3Nb_D_{022}	Al_3Ti (D_{022})	D_{022}	tI8	(139,I4/mmm)	$(Al, Nb)_3(Al, Nb)_1$
$AlNb_2$	sigma-CrFe (D_8b)	D_8b	tP30	(136, P4_2/mnm)	$(Al, Nb)_{10}(Nb)_4(Al, Nb)_{16}$
$AlRE_3$	Ni_3Sn (D_{019})	D_{019}	hP8	(194, P6_3/mmc)	$(Al)_1(Ce, La, Nd, Pr)_3$
AlR_2_C_{37}	Co_2Si (C_{37})	C_{37}	oP12	(62, Pnma)	$(Al)_1(Ce, Dy, Nd, Pr, Sm, Tb)_2$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
AlR_OP_{16}	DyAl	-	oP16	(57, Pbcm)	$(Al)_1(Dy, Nd, Pr, Sm, Tb)_1$
AlR_OS_{16}	AlCe	-	oS16	(63, Cmcm)	$(Al)_1(La, Pr)_1$
$AlZr_2\_B_{82}$	Ni_2In (B_{82})	B_{82}	hP6	(194, P6_3/mmc)	$(Al, VA)_1(VA, Zr)_2$
MB_B_{33}	CrB (B_{33})	B_{33}	oS8	(63, Cmcm)	$(Al, Nb)_1(B)_1$
$REGa\_B_{33}$	CrB (B_{33})	B_{33}	oS8	(63, Cmcm)	$(Ce, Dy, La, Nd, Pr, Tb, Zr)_1(Ga)_1$
$AlZr\_B_{33}$	CrB (B_{33})	B_{33}	oS8	(63, Cmcm)	$(Al)_1(Zr)_1$
B_4RE	UB_4	-	tP20	(127,P4/mbm)	$(B)_4(Ce, Dy, La, Nd, Pr, Sm, Tb)_1$
B_5RE_2	Pr_2B_5	-	mS56	(15, C2/c)	$(B)_5(Ce, Nd, Pr)_2$
$B_{66}RE$	YB_{66}	-	cF1936	(226, Fm-3c)	$(B)_{66}(Dy, Nd, Sm, Tb)_1$
B_6RE	CaB_6 (D_{21})	D_{21}	cP7	(221, Pm-3m)	$(B)_6(B, Ce, Dy, La, Nd, Pr, Sm, Tb)_1$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
BCC_A_2	Body-Centered Cubic (W, A_2, bcc)	A_2	cI2	(229, Im-3m)	(Al, B, Ce, Co, Cu, Dy, Fe, Ga, La, Nb, Nd, Pr, Sm, Tb, $Zr)_1(VA)_3$
BCC_B_2	CsCl (B_2)	B_2	cP2	(221, Pm-3m)	(Al, B, Ce, Co, Cu, Dy, Fe, Ga, La, Nb, Nd, Pr, Sm, Tb, $Zr)_1$(Al, B, Ce, Co, Cu, Dy, Fe, Ga, La, Nb, Nd, Pr, Sm, Tb, $Zr)_1(VA)_6$
BETA_RHOMBO_B	beta-B (R-$_{105}$)	-	hR105	(166, R-3m)	$(B)_{93}$(B, Cu, $Nb)_{12}$
C_{14}_LaVES	$MgZn_2$ Hexagonal Laves (C_{14})	C_{14}	hP12	(194, P6_3/mmc)	(Al, Co, Fe, Nb, $Zr)_2$(Al, Co, Fe, Nb, $Zr)_1$
C_{15}_LaVES	Cu_2Mg Cubic Laves (C_{15})	C_{15}	cF24	(227, Fd-3m)	(Al, Co, Dy, Fe, La, Nb, Nd, $Zr)_2$(Al, Ce, Co, Dy, Fe, La, Nb, Nd, Pr, Sm, Tb, $Zr)_1$
EPSILON	Hexagonal Close Packed (Mg, A_3, hcp)	A_3	hP2	(194, P6_3/mmc)	$(Cu)_1$
Fe_2B	Khatyrkite (Al_2Cu, C_{16})	C_{16}	tI12	(140, I4/mcm)	(Al, Co, $Fe)_2(B)_1$
C_{16}_THETA	Khatyrkite (Al_2Cu, C_{16})	C_{16}	tI12	(140, I4/mcm)	(Al, $Zr)_2$(Al, Co, Cu, Fe, $Ga)_1$
Co_3Nb_C_{36}	$MgNi_2$ Hexagonal Laves (C_{36})	C_{36}	hP24	(194, P6_3/mmc)	(Co, $Cu)_3(Nb)_1$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
$Ce_{24}Co_{11}$	$Ce_{24}Co_{11}$	-	hP70	(186, P6_3mc)	$(Ce)_{24}(Co, Fe)_{11}$
Ce_3Ga_2	Ba_5Si_3	-	tP32	(130, P4/ncc)	$(Ce)_3(Ga)_2$
CeMENTITE_D_{011}	Cementite (Fe_3C, D_{011})	D_{011}	oP16	(62, Pnma)	$(Co, Fe)_3(B)_1$
$REGa_6$	Ga_6Pu	-	tP14	(125, P4/nbm)	$(Ce, Dy, La, Nd, Pr, Tb)_1(Ga)_6$
$CeGa_6$_HT	Unknown Structure	-	-	-	$(Ce)_1(Ga)_6$
FeB	FeB (B_{27})	B_{27}	oP8	(62, Pnma)	$(Co, Fe, Zr)_1(B)_1$
CuRE_B_{27}	FeB (B_{27})	B_{27}	oP8	(62, Pnma)	$(Cu)_1(Ce, La, Nd, Pr, Sm)_1$
$Co_{11}Zr_2$	($Co_{11}Hf_2$)	-	oP*	(50, Pban)	$(Co, Fe)_{11}(Zr)_2$
$Co_{13}La$_D_{23}	$NaZn_{13}$ (D_{23})	D_{23}	cF112	(226, Fm-3c)	$(Co, Fe)_{13}(La)_1$
Co_3La_2	La_2Ni_3	-	oS20	(64, Cmce)	$(Co)_3(La, Nd)_2$
$Co_{23}La_{27}$	$LaCo_{0.85}$	-	mS8	(12, C2/m)	$(Co)_{23}(La)_{27}$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
Co_3Nd_2_HT	Unknown Structure	-	-	-	$(Co)_3(Nd)_2$
Co_3RE_4	Co_3Ho_4	-	hP22	(176, P6_3/m)	$(Co, Fe)_3(Nd, Pr, Tb)_4$
Co_2Nd_5	Mn_5C_2 (Fe_5C_2 Hagg carbide)	-	mS28	(15, C2/c)	$(Co)_2(Nd, Pr, Sm)_5$
Co_5Sm	Cu_7Tb	-	hP8	(191, P6/mmm)	$(Co, Fe)_5(Sm)_1$
Co_7Nb_2	Unknown Structure	-	-	-	$(Co)_7(Nb)_2$
Co_7Zr_2	Ni_7Zr_2	-	mS36	(12, C2/m)	$(Co)_7(Zr)_2$
$CoRE_3$_D_{011}	Cementite (Fe_3C, D_{011})	D_{011}	oP16	(62, Pnma)	$(Co)_1(Dy, La, Nd, Pr, Sm, Tb)_3$
$Cu_{10}Zr_7$	$Ni_{10}Zr_7$	-	oS68	(64, Cmce)	$(Cu)_{10}(Zr)_7$
M_2RE	KHg_2	-	oI12	(74, Imma)	$(Cu)_2(Ce, Dy, La, Nd, Pr, Sm, Tb)_1$
$Cu_{37}La_3$	$NaZn_{13}$ (D_{23})	D_{23}	cF112	(226, Fm-3c)	$(Cu)_{37}(La)_3$
Cu_4Ce	Unknown Structure	-	oP20	-	$(Al, Cu)_4(Ce)_1$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
Cu_4La_1	Cu_4La	-	tI90	(119, I-4m2)	$(Cu)_4(La)_1$
Cu_4Nd	Unknown Structure	-	o**	-	$(Cu)_4(Nd, Sm)_1$
Cu_4Pr	Unknown Structure	-	o**	-	$(Cu)_4(Pr)_1$
$Cu_{51}Zr_{14}$	$Ag_{51}Gd_{14}$	-	hP68	(175, P6/m)	$(Cu)_{51}(Zr)_{14}$
Cu_6Ce	Copper(II) Azide [Cu $(N_3)_2$]	-	oP28	(62, Pnma)	$(Cu)_6(Ce)_1$
Cu_6La_1_LT	Cu_6La	-	mP28	(14, P2_1/c)	$(Cu)_6(La)_1$
Cu_6Pr	Cu_6La	-	mP28	(14, P2_1/c)	$(Cu)_6(Pr)_1$
Cu_6R	$CeCu_6$	-	oP28	(62, Pnma)	$(Cu)_6(La, Nd, Sm)_1$
Cu_7Dy	Cu_7Tb	-	hP8	(191,P6/mmm)	$(Cu)_7(Dy)_1$
Cu_7RE_2	Unknown Structure	-	-	-	$(Cu)_7(Dy, Nd)_2$
Cu_8Zr_3	Cu_8Hf_3	-	oP44	(62, Pnma)	$(Cu)_8(Zr)_3$
Cu_9Ga_4_1	gamma-brass (Cu_9Al_4, D_{83})	D_{83}	cP52	(215, P-43m)	$(Cu)_6(Cu, Ga)_3(Cu, Ga)_3(Ga)_1$
Cu_9Ga_4_2	$Cu_{8.2}Ga_{4.8}$	-	cP52	-	$(Cu)_3(Cu, VA)_3(Cu, Ga)_3(Ga)_4$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
Cu_9Ga_4_3	$Cu_{7.15}Ga_{5.85}$	-	cP52	-	$(Cu, VA)_6(Cu, Ga)_3(Ga)_4$
$CuGa_2$	$FeSi_2$-h	-	tP3	(123, P4/mmm)	$(Cu)_1(Ga)_2$
CuGa_THETA	Unknown Structure	-	-	-	$(Cu)_7(Ga)_2$
CuNd_H	Unknown Structure	-	-	-	$(Cu)_1(Nd)_1$
CuR_B_2	CsCl (B_2)	B_2	cP2	(221, Pm-3m)	$(Cu)_1(Ce, Dy, La, Nd, Pr, Sm, Tb, Zr)_1$
$CuZr_2$_C_{11}B	$MoSi_2$ (C_{11}b)	C_{11}b	tI6	(139,I4/mmm)	$(Al, Cu)_1(Al, Zr)_2$
DHCP	alpha-La (A_3')	A_3'	hP4	(194, P6_3/mmc)	$(Al, Ce, Cu, Dy, Ga, La, Nd, Pr, Sm, Tb, Zr)_1$
REB_{12}	UB_{12} (D_2f)	D_2f	cF52	(225, Fm-3m)	$(Dy, Tb)_1(B)_{12}$
$CeCo_3B_2$	$CeCo_3B_2$	-	hP6	(191,P6/mmm)	$(Ce, Dy, Sm, Tb)_1(Co)_3(B)_2$
DIAMONd_A_4	Diamond (A_4)	A_4	cF8	(227, Fd-3m)	$(B, Ga)_1$
Dy_3FeB_7	Er_3CrB_7	-	oS44	(63, Cmcm)	$(Dy, Tb)_3(Fe)_1(B)_7$
$DyFeB_4$	$YCrB_4$	-	oP24	(55, Pbam)	$(Dy, Sm, Tb)_1(Co, Fe)_1(B)_4$
Dy_3Ga_5	Tm_3Ga_5	-	oP32	(62, Pnma)	$(Dy)_3(Ga)_5$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
FCC_A_1	Face-Centered Cubic (Cu, A_1, fcc)	A_1	cF4	(225, Fm-3m)	(Al, B, Ce, Co, Cu, Dy, Fe, Ga, La, Nb, Nd, Pr, Sm, Tb, Zr)$_1$(VA)$_1$
RE_2M_{17}_HEX	$Ni_{17}Th_2$	-	hP38	(194, P6_3/mmc)	(Al, Co, Fe)$_{17}$(Dy, Nd, Pr, Sm, Tb)$_2$
$Fe_1Nb_1B_1$_C_{22}	Revised Fe_2P (C_{22})	C_{22}(II)	hP9	(189, P-62m)	(Fe)$_1$(Nb)$_1$(B)$_1$
$Fe_{13}RE_6X$	$Fe_{13}Pr_6Ge$	-	tI80	(140, I4/mcm)	(Fe)$_{13}$(Nd, Pr)$_6$(Cu, Ga)$_1$
$Fe_3Nb_3B_4$	Unknown Structure	-	-	-	(Fe)$_3$(Nb)$_3$(B)$_4$
RE_2M_{17}	$Zn_{17}Th_2$	-	hR57	(166, R-3m)	(Al, Co, Fe)$_{17}$(Ce, Dy, La, Nd, Pr, Sm, Tb)$_2$
$Al_{10}Ce_2M_7$	$Zn_{17}Th_2$	-	hR57	(166, R-3m)	(Al, Cu, Fe)$_{17}$(Ce)$_5$
$Fe_{17}RE_5$	Nd_5Fe_{17}	-	hP264	(194, P6_3/mmc)	(Fe)$_{17}$(Ce, La, Nd, Pr, Sm)$_5$
$Fe_{23}R_6$	Th_6Mn_{23} (D_8a)	D_8a	cF116	(225, Fm-3m)	(Co, Fe)$_{23}$(Dy, Tb, Zr)$_6$
Fe_3Ga_4	Fe_3Ga_4	-	mS42	(12, C2/m)	(Fe)$_3$(Ga)$_4$
Fe_6Ga_5	Fe_6Ge_5	-	mS44	(12, C2/m)	(Fe)$_6$(Ga)$_5$
$FeGa_3$	In_3Ir	-	tP16	(136, P4_2/mnm)	(Co, Fe)$_1$(Ga)$_3$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
Fe_3RE	Ni_3Pu	-	hR12	(166, R-3m)	(Ce, Dy, Nd, Pr, Sm, Tb)$_1$(Co, Fe)$_3$
$FeZr_3$	Re_3B	-	oS16	(63, Cmcm)	(Co, Fe)$_1$(Zr)$_3$
Ga_3RE	Bogdanovite (Cu_3Au, L_{12})	L_{12}	cP4	(221, Pm-3m)	(Ga)$_3$(Dy, Tb)$_1$
Ga_4La	Ga_4La	-	oP20	(47, Pmmm)	(Ga)$_4$(La, Pr)$_1$
GaMMA_D_{83}	gamma-brass (Cu_9Al_4, D_{83})	D_{83}	cP52	(215, P-43m)	(Al, Fe)$_4$(Al, Cu)$_1$(Cu, Fe)$_8$
GaMMA_H	gamma-brass (Cu_5Zn_8, D_{82})	D_{82}	cI52	(217, I-43m)	(Al)$_4$(Al, Cu)$_1$(Cu, Fe)$_8$
$GaPr_2$_C_{37}	Co_2Si (C_{37})	C_{37}	oP12	(62, Pnma)	(Ga)$_1$(Pr)$_2$
HCP_A_3	Hexagonal Close Packed (Mg, A_3, hcp)	A_3	hP2	(194, P6_3/mmc)	(Al, Ce, Co, Cu, Dy, Fe, Ga, La, Nb, Nd, Pr, Sm, Tb, Zr)$_2$(VA)$_1$
La_1B_9	Unknown Structure	-	-	-	(La)$_1$(B)$_9$
L_{12}_FCC	Bogdanovite(Cu_3Au, L_{12})	L_{12}	cP4	(221, Pm-3m)	(Al, Ga)$_1$(Ce, Fe, La, Nd, Pr, Zr)$_3$
$LaCo_2B_3$	Unknown Structure	-	-	-	(La)$_1$(Co)$_2$(B)$_3$
LIQUID	Liquid	-	-	-	(Al, Al_2Sm, B, Ce, Co, Cu, Dy, Fe, Ga, La, Nb, Nd, Pr, Sm, Tb, Zr)$_1$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
M_3B_4_D_7B	Ta_3B_4 (D_7b)	D_7b	oI14	(71, Immm)	$(B)_4(Al, Nb)_3$
M_5R_$C_{15}B$	$AuBe_5$ ($C_{15}b$)	$C_{15}b$	cF24	(216, F-43m)	$(Cu)_5(Zr)_1$
Ga_3Tb_5_D_8L	Cr_5B_3 (D_8l)	D_8l	tI32	(140, I4/mcm)	$(Ga)_3(Dy, Tb)_5$
Ga_2Zr_1	Ga_2Zr	-	oS12	(65, Cmmm)	$(Ga)_2(Zr)_1$
Ga_5Zr_3	Pd_5Pu_3	-	oS32	(63, Cmcm)	$(Ga)_5(Zr)_3$
GaZr_BG	MoB (Bg)	Bg	tI16	(141, I4_1/amd)	$(Ga)_1(Zr)_1$
Nb_2B_3	V_2B_3	-	oS20	(63, Cmcm)	$(Al, Nb)_2(B)_3$
Nb_5B_6	V_5B_6	-	oS22	(65, Cmmm)	$(Al, Nb)_5(B)_6$
$Nd_3Co_{13}B_2$	$Nd_3Ni_{13}B_2$	-	hP18	(191,P6/mmm)	$(Nd, Sm)_3(Co)_{13}(B)_2$
$Nd_2Co_5B_2$	$Ce_2Co_5B_2$	-	hP36	(194, P6_3/mmc)	$(Nd, Pr, Tb)_2(Co)_5(B)_2$
$Nd_2Co_5B_3$	*	-	oS*	(68, Ccce)	$(Nd, Pr)_2(Co)_5(B)_3$
ORTHORHOMBIC_Ga	alpha-Ga (A_{11})	A_{11}	oS8	(64, Cmce)	$(Ga)_1$
RE_2Co_7	Co_7Gd_2	-	hR18	(166, R-3m)	$(Ce, Dy, La, Nd, Pr, Sm, Tb)_2(Co, Fe)_7$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
$Ce_2Co_7B_3$	$Ce_2Co_7B_3$	-	hP24	(191,P6/mmm)	(Ce, Dy, Nd, Pr, Sm, Tb)$_2$(Co)$_7$(B)$_3$
RE_5Co_{19}	Ce_5Co_{19}	-	hR24	(166, R-3m)	(Ce, La, Nd, Pr, Sm)$_5$(Co, Fe)$_{19}$
$RECo_5$_D_2D	$CaCu_5$ (D_2d)	D_2d	hP6	(191,P6/mmm)	(Ce, Dy, La, Nd, Pr, Sm, Tb)$_1$(Co, Fe)$_5$
$RE_3Co_{11}B_4$	$Ce_3Co_{11}B_4$	-	hP19	(191,P6/mmm)	(Ce, Dy, Nd, Pr, Sm, Tb)$_3$(Co, Fe)$_{11}$(B)$_4$
$RECo_2B_2$	$ThCr_2Si_2$	-	tI10	(139,I4/mmm)	(Dy, La, Pr, Sm, Tb)$_1$(Co, Fe)$_2$(B)$_2$
$RECo_4B$	$CeCo_4B$	-	hP12	(191,P6/mmm)	(Ce, Dy, La, Nd, Pr, Sm, Tb)$_1$(Co, Fe)$_4$(B)$_1$
$RECo_{12}B_6$	$SrNi_{12}B_6$	-	hR19	(166, R-3m)	(Ce, Dy, La, Nd, Pr, Sm, Tb)$_1$(Co, Fe)$_{12}$(B)$_6$
RHOMB_C_{19}	alpha-Sm (C_{19})	C_{19}	hR3	(166, R-3m)	(Al, Ce, Co, Dy, Nd, Sm)$_1$
RM_5	$CaCu_5$ (D_2d)	D_2d	hP6	(191,P6/mmm)	(Dy, La, Nd, Sm, Tb)$_1$(Cu)$_5$
Cu_5Ce	$CaCu_5$ (D_2d)	D_2d	hP6	(191,P6/mmm)	(Al, Cu)$_5$(Ce)$_1$
Cu_5Pr	$CaCu_5$ (D_2d)	D_2d	hP6	(191,P6/mmm)	(Cu)$_5$(Pr)$_1$
MU_D_{85}	Fe_7W_6 (D_{85}) mu-phase	D_{85}	hR13	(166, R-3m)	(Co, Fe, Nb)$_1$(Nb)$_4$(Co, Fe, Nb)$_2$(Co, Fe, Nb)$_6$
M_3B_2_D_5A	Si_2U_3 (D_5a)	D_5a	tP10	(127,P4/mbm)	(B)$_2$(Fe, Nb)$_3$

Phase Name	Prototype	Strukturbericht	Pearson Symbol	Space Group	Formula Unit
Ga_2Zr_3_D_5A	Si_2U_3 (D_5a)	D_5a	tP10	(127,P4/mbm)	$(Ga)_2(Zr)_3$
T_1	$Nd_2Fe_{14}B$	-	tP68	(136, P4_2/mnm)	$(Al, Co, Fe)_{14}(Ce, Dy, La, Nd, Pr, Sm, Tb)_2(B)_1$
$CeCo_4B_4$	$CeCo_4B_4$	-	tP18	(137, P4_2/nmc)	$(Co)_4(Ce, Dy, Nd, Pr, Sm, Tb)_1(B)_4$
T_2	$Nd_{19}Fe_{68}B_{68}$	-	tP310	(86, P4_2/n)	$(Fe)_{68}(Ce, Dy, La, Nd, Pr, Sm, Tb)_{19}(B)_{68}$
T_3	$Pr_5Co_2B_6$	-	hR39	(166, R-3m)	$(Co, Fe)_2(Ce, Dy, La, Nd, Pr, Sm, Tb)_5(B)_6$
Tb_4CoB_{13}	Er_4NiB_{13}	-	tP36	(128, P4/mnc)	$(Tb)_4(Co)_1(B)_{13}$
ZrB_{12}	UB_{12} (D_2f)	D_2f	cF52	(225, Fm-3m)	$(B)_{12}(Zr)_1$
GaS	Gas	-	-	-	$(Al, Al_1Cu_1, Al_2, B, B_2, Ce, Cu, Cu_2, Dy, Fe, Fe_2, Ga, Ga_2, La, Nb, Nd, Pr, Sm, Tb , Zr, Zr_2)_1$